NORTH SOUTH UNIVERSITY · DEPT. OF ELECTRICAL & COMPUTER ENGINEERING (ECE) · CSE499 CAPSTONE

A Voice-Based AI Framework for Medical Pre-Screening of Bangla & Regional Bangladeshi Speech

Intelligent Symptom Capture · Risk Assessment · Clinical Documentation · EHR Integration

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1 Introduction, Objective & Novelty

Introduction

- In Bangladeshi clinics, symptom history is taken verbally in **Bangla, Banglish or dialect** under heavy time pressure — details are lost before consultation.
- A raw transcript alone is **neither complete nor structured**, so it cannot serve as a clinical record.
- Voice-first intake suits **elderly, low-literacy and non-technical** patients better than typed forms.

Objective

- Capture symptoms **naturally by voice** (typing fallback) in Bangla / Banglish / dialect.
- Produce a **verified, structured** summary; fill gaps through spoken follow-up.
- Support triage, **keep the doctor in the loop**, and preserve the patient's own words.
- Generate a doctor-ready **EHR / FHIR** record — **assist, never diagnose**.

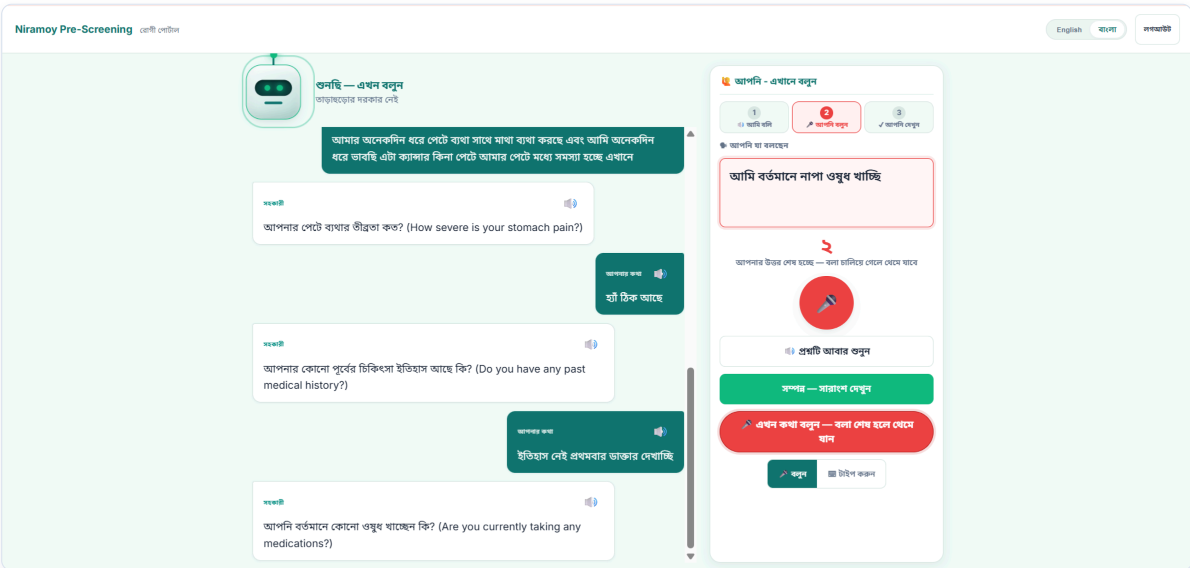
Problem & Novelty

- Dialects, code-switching, medical terms & clinic noise make single-pass Bangla ASR **unreliable**.
- **Novelty:** a stateful, conversational pre-screener that **verifies & fills gaps**, not just one transcript.
- Rule-based red-flag safety, **explainable** risk, and a dialect-aware, **three-portal, human-in-the-loop** workflow.

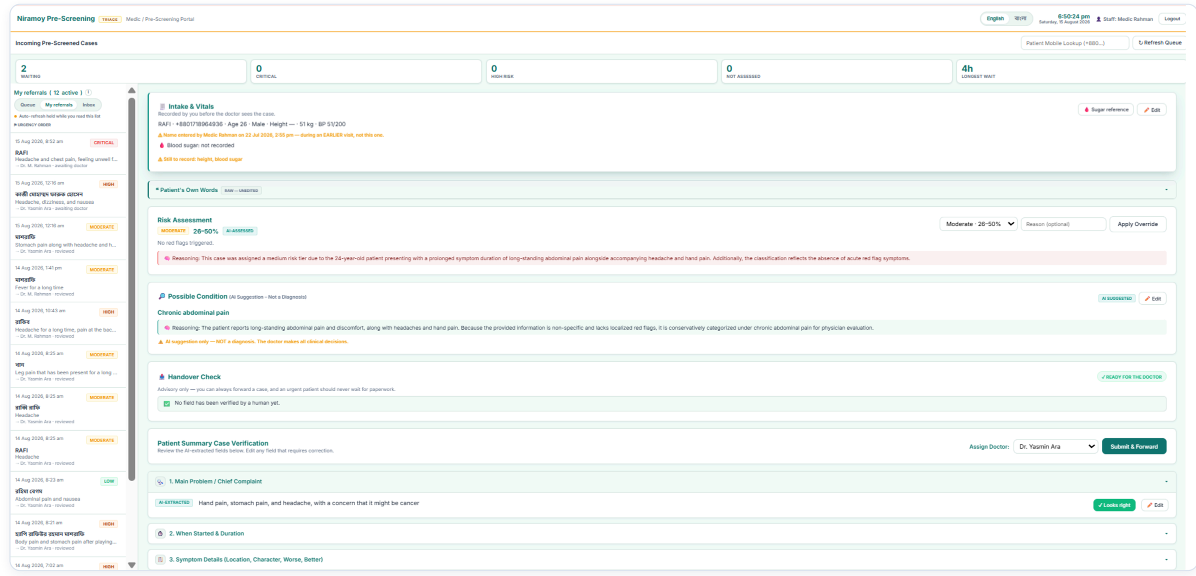
2 Methodology — From Spoken Symptoms to a Doctor-Ready Record

1. **Patient Voice** — Patient speaks symptoms in Bangla / dialect (typing fallback).
2. **Speech Recognition** — Web Speech API (bn-BD) transcribes; **raw words stored unchanged.**
3. **Confirm & Correct** — Cleaned in a separate field, shown back to confirm.
4. **Structured Capture** — LLM builds a 10-field summary; follow-ups loop until complete.
5. **Assessment & Triage** — Risk Low → Critical with rule-based **red-flag check** + reason.
6. **Doctor Review** — Doctor accepts, overrides or prescribes — decision stays human.
7. **Clinical Record / EHR** — No-diagnosis report stored & exported: **FHIR / PDF / DOCX**.

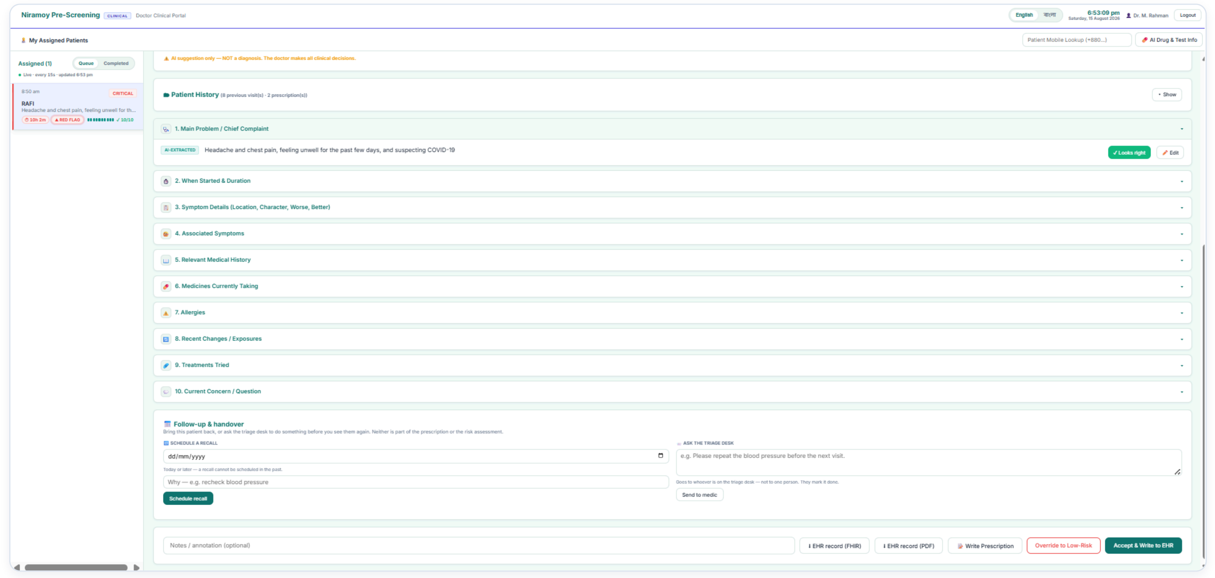
3 System Features — Three Portals, Clinical Record & Speech-Recognition Performance



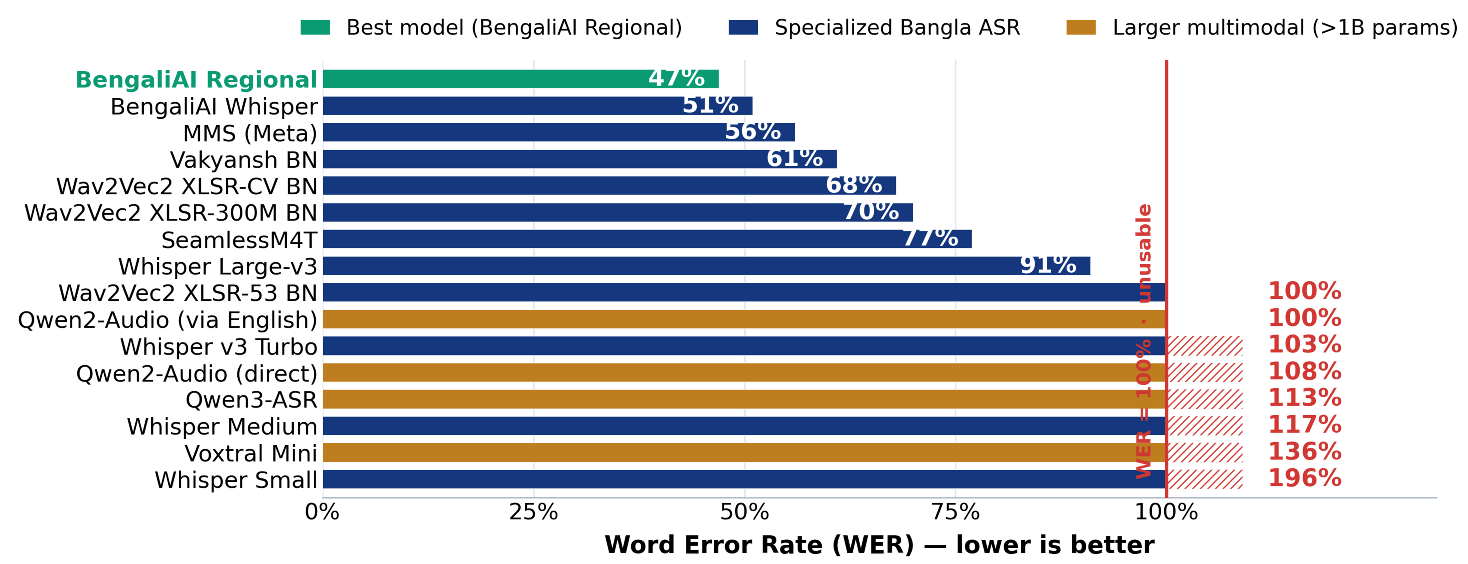
Patient Portal — Voice-First Capture
Patient answers spoken follow-ups in Bangla by voice; exact words are kept and each field confirmed.



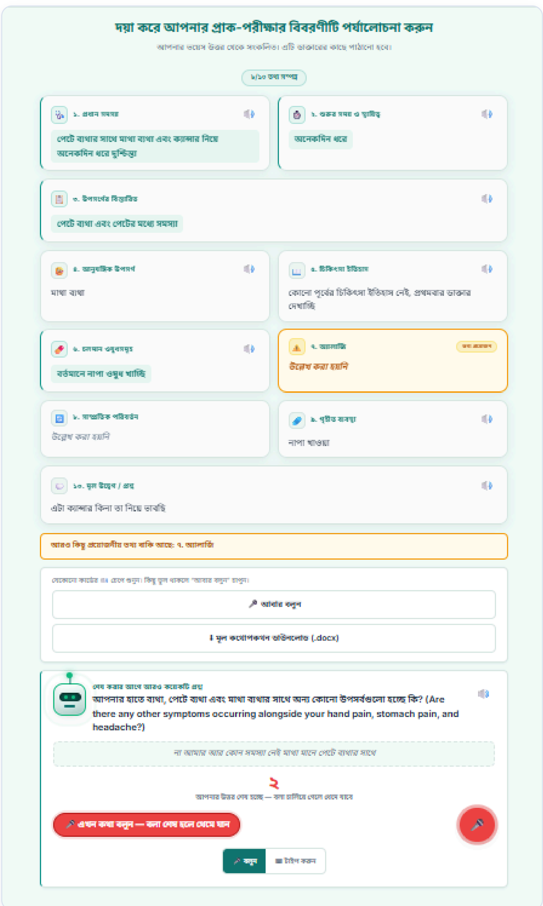
Medic Portal — Triage & Verification
Cases ranked by risk with the AI's reasoning; the medic records vitals and verifies fields before a doctor.



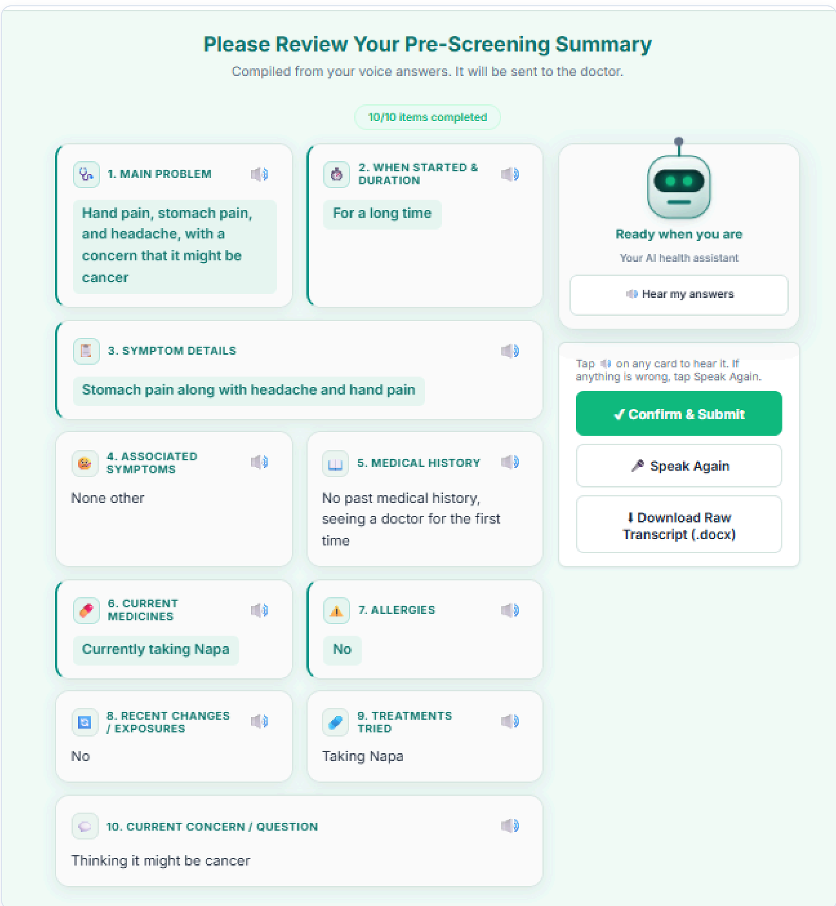
Doctor Portal — Review & Decision
Safety panel (risk tier, red flags, explanation) + history; the doctor accepts, overrides or prescribes.



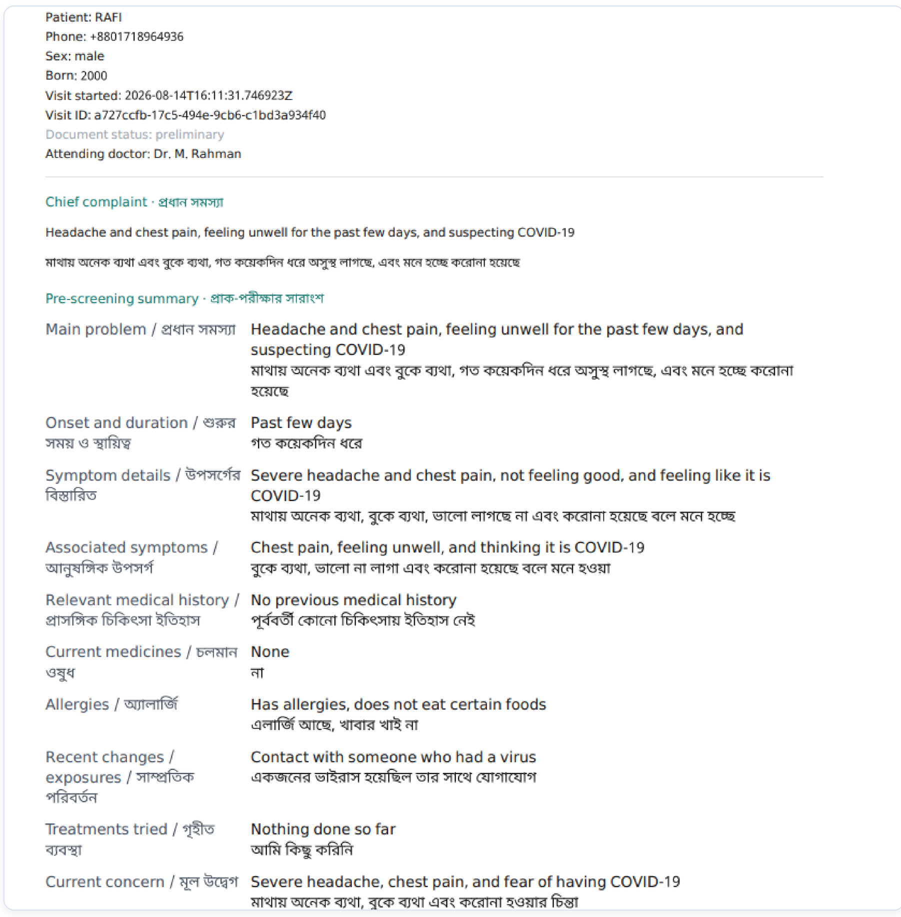
Speech-Recognition Benchmark (WER)
16 Bangla-dialect ASR systems on multi-dialect medical speech; bars past the red wall are unusable.



Patient — Bangla Review



Patient — English Review & Confirm


Patient: RAFI
Phone: +8801718964936
Sex: male
Born: 2000
Visit started: 2026-08-14T16:11:31.746923Z
Visit ID: a727ccfb-17c5-494e-9cb6-c1bd3a934f40
Document status: preliminary
Attending doctor: Dr. M. Rahman

Chief complaint
Headache and chest pain, feeling unwell for the past few days, and suspecting COVID-19

Pre-screening summary

Main problem	Headache and chest pain, feeling unwell for the past few days, and suspecting COVID-19
Onset and duration	Past few days
Symptom details	Severe headache and chest pain, not feeling good, and feeling like it is COVID-19
Associated symptoms	Chest pain, feeling unwell, and thinking it is COVID-19
Relevant medical history	No previous medical history
Current medicines	None
Allergies	Has allergies, does not eat certain foods
Recent changes / exposures	Contact with someone who had a virus
Treatments tried	Nothing done so far
Current concern	Severe headache, chest pain, and fear of having COVID-19

Clinical Record — EHR / FHIR Output

Patient Review & the Shared Clinical Record

Bangla review: the 10 captured fields are shown back fully bilingual, so non-English-speaking and low-literacy patients can verify and correct their own record.

English review & confirm: the same confirmation screen in English; the patient approves before submission, improving data quality with informed control over their own record.

EHR / FHIR record: a structured, bilingual, no-diagnosis pre-screening document stored in the EHR and exported as **HL7 FHIR R4, PDF & DOCX** — a doctor-ready record for the consultation.

Speech-Recognition Study & Data

~**4.7 h** multi-dialect audio (16 kHz mono) collected & preprocessed; 16 ASR systems scored by **WER / CER / BLEU** per dialect. The best model still errs ~47%; larger >1B models did not help.

16	47%	8
ASR models tested	best WER	failed (≥100%)

Honesty note: the WER study is an **offline research benchmark**; the live prototype uses the browser **Web Speech API**, not these models.

Expected Impact

Faster, more complete intake and fewer missed details — the doctor receives a structured record they can trust, **without added clinical risk or replacing judgment**. One shared record links all three portals.

4 Technology Stack · Limitations & Future Work · Conclusion

Technology Stack

BACKEND & FRONTEND
Python · FastAPI · REST + WebSocket · HTML · CSS · JS · 3 portals

SPEECH · AI / LLM
Web Speech API (bn-BD) · edge-tts · Gemini · Groq · OpenRouter · + fallback

CLINICAL DATA · INFRA
SQLite + Alembic · HL7 FHIR R4 · fpdf2 · python-docx · pytest

Design: modular services · swappable AI providers · CPU-only · Windows + Arch Linux.

Limitations & Future Work

LIMITATIONS

- Live STT uses the cloud browser API; dialect recognition stays error-prone.
- Login / OTP & encryption are prototype-stage, not production security.
- Extraction accuracy not yet formally measured (free-tier LLMs).

FUTURE WORK

- On-device **quantised** STT & summary — offline & private.
- Hands-free follow-up; LoRA dialect ASR fine-tuning.
- Formal extraction evaluation; real auth, encryption & SMS OTP.

Conclusion

A complete **voice-first, safety-first** prototype turns spoken Bangla / dialect symptoms into a structured, doctor-ready **EHR / FHIR** record across three portals. It **reduces intake burden and keeps clinical judgment with humans** — it assists, and **never diagnoses**.

Key Contributions

- End-to-end **three-portal** prototype (voice → record → EHR/FHIR).
- **16-model** Bangla-dialect ASR benchmark.
- Red-flag safety with **explainable**, human-in-the-loop risk.